

## IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

### PROJECT 50

# Delegate CloudFormation StackSets and Deploy an Organization-Wide Event Forwarder Role

AWS Organizations | CloudFormation StackSets | IAM | EventBridge

|              |                  |
|--------------|------------------|
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| ENVIRONMENT  | Personal AWS Lab |
| VERSION      | v1.0.0           |
| RELEASE DATE | August 19, 2026  |
| STATUS       | Complete         |

## Executive Summary

Enabled trusted access for **AWS CloudFormation StackSets** from the AWS Organizations management account, registered **SecurityOperations** as delegated administrator, and used SecurityOperations to create a **service-managed StackSet** named **SecOpsEventForwarderRole**. The StackSet targets the AWS Organization and deploys an IAM role named **SecurityEventForwarder** that Amazon EventBridge can assume to publish to the central **SecurityAutomation** event buses created in Project 49.

### EVIDENCE CORRECTION

The written lab summary says 10 maximum concurrent accounts and parallel deployment. The supplied review screenshot shows 100 maximum concurrent accounts, Failure tolerance 99, Region concurrency SEQUENTIAL, and STRICT\_FAILURE\_TOLERANCE. The project records the screenshot-grounded settings. Because only us-east-1 is selected, region concurrency does not affect this run.

## Business Problem

Project 49 created centralized SecurityAutomation EventBridge buses, but member accounts still need a consistent IAM role for cross-account event forwarding. Manually creating that role in every account would increase drift and maintenance. StackSets provides repeatable organization-scale deployment and automatic coverage as accounts are added.

## Objectives

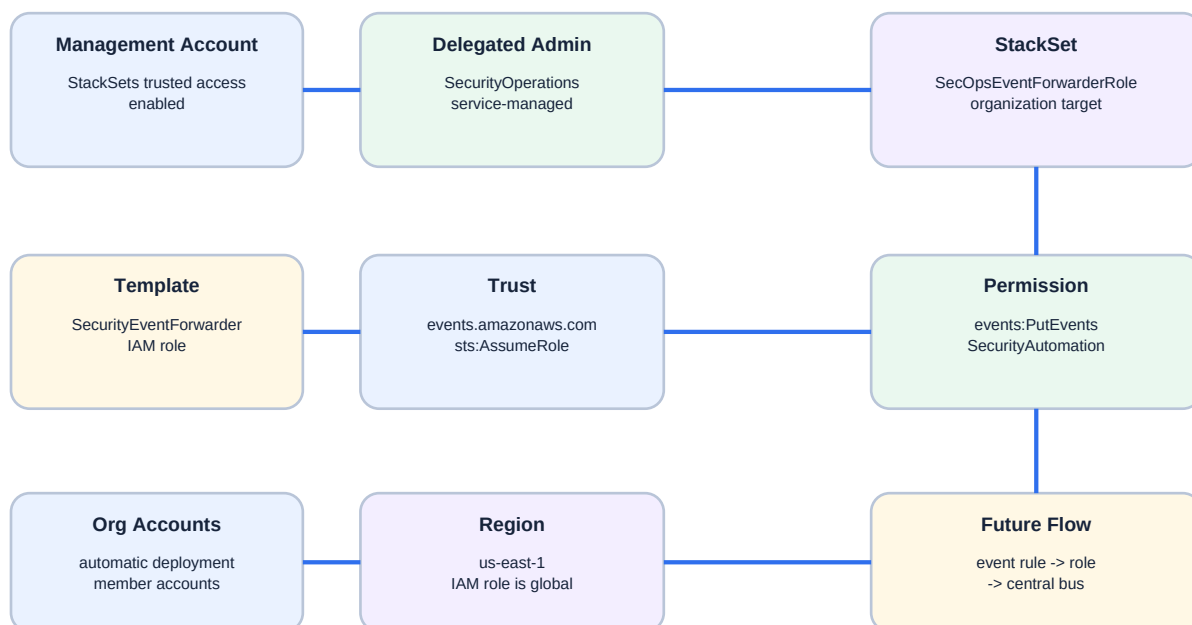
- Enable trusted access between CloudFormation StackSets and AWS Organizations.
- Delegate StackSets administration to SecurityOperations.
- Use service-managed permissions for organization-scale automatic deployment.
- Deploy the SecurityEventForwarder IAM role to member accounts.
- Allow Amazon EventBridge to assume the forwarding role.
- Allow the role to publish to SecurityAutomation event buses in us-east-1 and us-west-2.
- Target the organization root and enable automatic deployment.
- Deploy in us-east-1 because the resource being created is an IAM role.
- Confirm the StackSet is active.

## Environment

| Management activity     | Enable CloudFormation StackSets trusted access |
|-------------------------|------------------------------------------------|
| Delegated administrator | SecurityOperations                             |

| StackSet permission model   | SERVICE_MANAGED                                        |
|-----------------------------|--------------------------------------------------------|
| StackSet name               | SecOpsEventForwarderRole                               |
| Description                 | Role to forward events to the security operations team |
| Target                      | AWS Organization root                                  |
| Automatic deployment        | Activated                                              |
| Account removal behavior    | Delete stacks                                          |
| Region                      | us-east-1                                              |
| Maximum concurrent accounts | 100 - evidence screenshot                              |
| Failure tolerance           | 99                                                     |
| Region concurrency          | SEQUENTIAL - evidence screenshot                       |
| Concurrency mode            | STRICT_FAILURE_TOLERANCE                               |
| Status                      | ACTIVE / successfully created                          |

## Architecture



StackSets distributes the EventBridge forwarding IAM role across the organization.

Figure 1 - Organization-wide EventBridge forwarding role deployment through CloudFormation StackSets.

## Implementation Summary

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1. Logged into the AWS Organizations management account.
2. Opened CloudFormation > StackSets and enabled trusted access.
3. Registered SecurityOperations as delegated administrator.
4. Logged out of the management account and signed into SecurityOperations.
5. Created a new StackSet using service-managed permissions.
6. Loaded the provided CloudFormation template for the EventBridge forwarding IAM role.
7. Provided the SecurityOperations account ID through SecurityEventBusAccountId.
8. Named the StackSet SecOpsEventForwarderRole.
9. Targeted the organization root and enabled automatic deployment.
10. Configured stacks to be deleted when accounts leave the target.
11. Selected us-east-1.
12. Recorded the review settings: max concurrent 100, failure tolerance 99, sequential region concurrency, strict failure tolerance.
13. Submitted the StackSet and confirmed ACTIVE status.

## CloudFormation Template

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```
AWSTemplateFormatVersion: '2010-09-09'
Description: "Creates IAM role for EventBridge cross-account event forwarding"

Parameters:
  SecurityEventBusAccountId:
    Type: String

Resources:
  EventBridgeForwardRole:
    Type: AWS::IAM::Role
    Properties:
      RoleName: SecurityEventForwarder
      Path: /SecurityOperations/
      AssumeRolePolicyDocument:
        Statement:
          - Effect: Allow
            Principal:
              Service: events.amazonaws.com
            Action: sts:AssumeRole
      Policies:
        - PolicyName: ForwardToEventBus
          PolicyDocument:
            Statement:
              - Effect: Allow
                Action: events:PutEvents
                Resource:
                  - arn:aws:events:us-east-1:<SecOpsAccountId>:event-bus/SecurityAutomation
                  - arn:aws:events:us-west-2:<SecOpsAccountId>:event-bus/SecurityAutomation
```

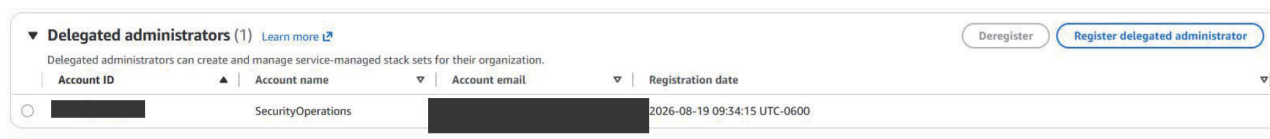
The template creates **SecurityEventForwarder** under the **/SecurityOperations/** path. EventBridge is trusted to assume the role, and the inline policy grants **events:PutEvents** only to the designated SecurityAutomation buses.

## Security Model

| Delegated administration | SecurityOperations manages service-managed StackSets | Reduces routine use of the management account.                |
|--------------------------|------------------------------------------------------|---------------------------------------------------------------|
| EventBridge trust        | events.amazonaws.com can assume the role             | Scopes role assumption to the EventBridge service.            |
| Forwarding permission    | events:PutEvents to SecurityAutomation only          | Limits the deployed role to event publication.                |
| Organization target      | Root + automatic deployment                          | Keeps the role present as member accounts are added.          |
| Single-region deployment | us-east-1                                            | Avoids unnecessary duplicate deployment of a global IAM role. |

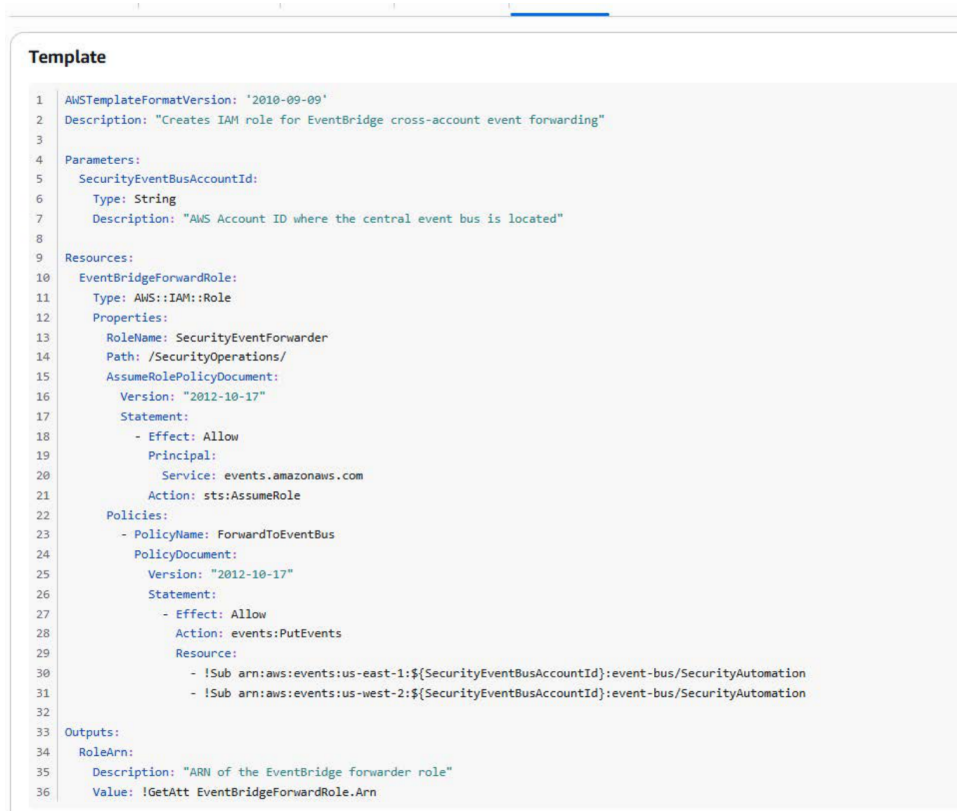
## Evidence

**Figure 2 - SecurityOperations Registered as Delegated Administrator**



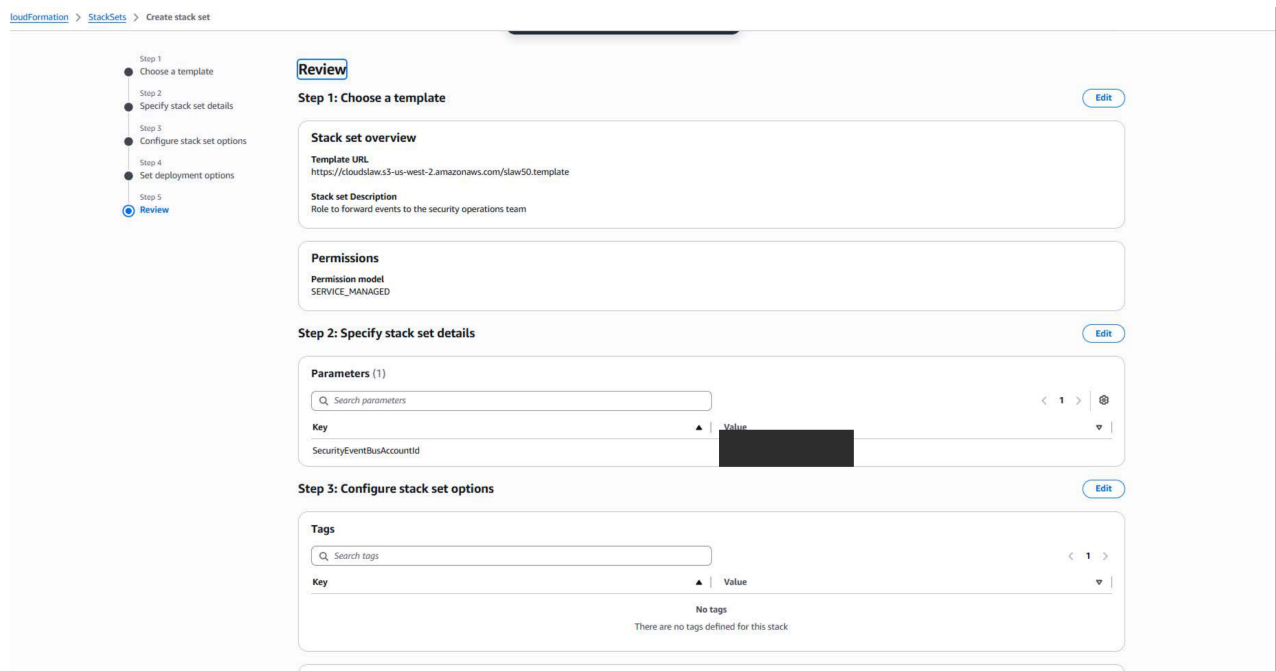
The delegated administrator evidence shows SecurityOperations registered for CloudFormation StackSets. Identifiers are redacted.

## Figure 3 - Event Forwarder CloudFormation Template



The template evidence shows the `SecurityEventBusAccountId` parameter, `SecurityEventForwarder` IAM role, `EventBridge` trust, and `events:PutEvents` permissions for `SecurityAutomation` in `us-east-1` and `us-west-2`.

## Figure 4 - StackSet Review - Template, Permissions, and Parameter



The review screenshot shows the template, `SERVICE_MANAGED` permission model, and `SecurityEventBusAccountId` parameter. The account ID was additionally redacted in this portfolio copy.

Figure 5 - StackSet Review - Organization Target and Deployment Controls

Deployment targets (1)

Q Search AWS OU IDs

< 1 >

AWS OU ID

██████████

Auto-deployment options

Automatic deployment

Activated

Retain stacks on account removal

Delete stacks

Stack set dependencies

Q Search stack set dependencies

< 1 >

Stack set name

Stack set ARN

No stack set dependencies

No stack set dependencies defined for this stack set

Regions

Q Search regions

< 1 >

Region

us-east-1

Deployment options

Maximum concurrent accounts

100

Failure tolerance

99

Region concurrency

SEQUENTIAL

Concurrency mode

STRICT\_FAILURE\_TOLERANCE

Cancel

Previous

Submit

The review shows the organization root target, automatic deployment, delete-stacks behavior, us-east-1, Maximum concurrent accounts 100, Failure tolerance 99, Region concurrency SEQUENTIAL, and STRICT\_FAILURE\_TOLERANCE. The root identifier was additionally redacted.

Figure 6 - SecOpsEventForwarderRole StackSet Active

SecOpsEventForwarderRole

Actions

Stack set info

Stack instances

Operations

Parameters

Template

Overview

Stack set status

ACTIVE

Stack set description

Role to forward events to the security operations team

Permission model

SERVICE\_MANAGED

Drift status

NOT\_CHECKED

Last drift check time

-

Stack set ID

SecOpsEventForwarderRole-██████████

Stack set ARN

████████████████████████████████████████████████████████████████████████████████

Stack set admin role ARN

████████████████████████████████████████████████████████████████████████████████

Managed execution

Inactive

Organizational unit IDs (1)

Q Search AWS OU IDs

< 1 >

AWS OU ID

r-egag

Auto-deployment options

Automatic deployment

Activated

Retain stacks on account removal

Delete stacks

Stack set dependencies

Q Search stack set dependencies

< 1 >

Stack set name

Stack set ARN

Edit automatic deployment

The final evidence shows SecOpsEventForwarderRole in ACTIVE status using SERVICE\_MANAGED permissions with automatic deployment activated.

# Security Analysis

StackSets is a privileged organization-wide administration capability, so delegated administration should remain tightly controlled.

Service-managed permissions simplify automatic deployment across the organization, but they also make StackSet administration a high-value security capability.

The deployed IAM role is narrowly scoped compared with administrator access: EventBridge can assume it, and its inline policy is limited to PutEvents against the designated buses.

Organization-root targeting ensures current and future member accounts receive the forwarding role automatically.

Deploying the IAM role in one region is sufficient because IAM roles are global, while the role policy can still reference EventBridge buses in multiple regions.

StackSet drift is an operational risk; future controls should monitor and protect shared security resources from manual modification.

## Lessons Learned

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CloudFormation StackSets can deploy shared security components consistently across many accounts.

Trusted access is required for service-managed organization deployments.

Delegated administration moves day-to-day StackSet operations away from the management account.

Automatic deployment makes StackSets useful for account bootstrap resources.

CloudFormation parameters allow one reusable template to target a centralized security account.

Captured review settings should be treated as authoritative evidence when written notes differ.

## Production Improvements

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- Restrict StackSet administration to a small number of SecurityOperations identities.
- Protect SecurityEventForwarder from modification or deletion with organization guardrails.
- Enable and review StackSet drift detection.
- Use a deployment pipeline and peer review for StackSet template changes.
- Monitor StackSet changes and delegated-administrator changes with CloudTrail and EventBridge.
- Use lower failure tolerance for higher-risk production deployments.
- Deploy future event-forwarding rules only to intended workload OUs when organization-wide scope is unnecessary.
- Validate the forwarding role in representative member accounts before production rule rollout.

## Skills Demonstrated

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|                |                          |                           |
|----------------|--------------------------|---------------------------|
|                |                          |                           |
| Trusted access | CloudFormation StackSets | Cross-account EventBridge |



| Delegated administration | Service-managed permissions | IAM role deployment      |
|--------------------------|-----------------------------|--------------------------|
| Automatic rollout        | CloudFormation parameters   | Central event forwarding |

## Resume Bullet

### RESUME

Delegated AWS CloudFormation StackSets administration to SecurityOperations and deployed a service-managed organization-wide EventBridge forwarding IAM role, creating repeatable cross-account access to centralized SecurityAutomation event buses for future security-event routing.

## STAR Interview Story

**Situation:** SecurityOperations had centralized SecurityAutomation EventBridge buses, but member accounts did not yet have a consistent IAM role for event forwarding.

**Task:** Deploy the forwarding role across the organization without manual per-account configuration or routine management-account administration.

**Action:** I enabled StackSets trusted access, delegated administration to SecurityOperations, created a service-managed organization-targeted StackSet from a parameterized template, enabled automatic deployment, and deployed the SecurityEventForwarder role through us-east-1.

**Result:** The organization now has a reusable StackSet bootstrap for the IAM role that future EventBridge rules will use to send security events to SecurityAutomation.

## Version History

| v1.0.0 | August 19, 2026 | Initial documentation. Enabled StackSets trusted access, delegated administration to SecurityOperations, created the service-managed SecOpsEventForwarderRole StackSet, targeted the organization, recorded evidence-grounded deployment settings, and embedded five evidence screenshots with additional identifier redaction. |
|--------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## References

- Cloud Security Lab a Week - Own (or PWN) the Org with CloudFormation StackSets.
- User-provided delegated administrator evidence.
- User-provided CloudFormation template evidence.
- User-provided StackSet review evidence.
- User-provided active StackSet evidence.
- Project 49 - EventBridge Security Automation and Broken RCP Replacement.